

Technology Stack

Date	03 October 2026
Team ID	17
Project Name	Fitbuddy-AI Fitness plan Generator
Department	Computer Technology
Institution / University	Madras Institute Of Technology / Anna University
Academic Year / Semester / Section	3 / 5 / Batch A
Faculty Mentor / Guide	Jayachitra V P
Team Leader	Bharath Jeyakkumar S
Sprint 1 Start / End	28-09-2026 / 28-09-2026
Sprint 2 Start / End	29-09-2026 / 30-09-2026
Sprint 3 Start / End	01-10-2026 / 02-10-2026
Team Roles	Abdul Wahith M — Backend & AI Integration; Kamalesh T — UI / Frontend Development; Bilu Besto H — Validation, Feedback & Reliability; Bharath Jeyakkumar S — Team Leader / Testing & Quality Assurance; Prasannavelan K — Database, Documentation & Demo Support
Maximum Marks	2 Marks

Layer	Technology	Purpose
Language	Python	Application implementation
Web framework	FastAPI	HTML routes and JSON APIs
Server	Uvicorn	ASGI application server
Templates	Jinja2	Server-rendered web pages
Validation	Pydantic	Input and API schema validation
ORM	SQLAlchemy	Database models and persistence
Database	SQLite	Local user/plan storage
AI	Google Gemini via google-genai	Workout, nutrition and revision generation

Layer	Technology	Purpose
Testing	pytest + FastAPI TestClient/ httpx	Automated application testing
Configuration	python-dotenv	Environment-based settings
Deployment	Dockerfile	Containerized execution

Team technology ownership

Member	Area
Abdul Wahith M	FastAPI architecture, AI integration and API
Kamalesh T	Jinja2 templates and CSS/UI
Bilu Besto H	Pydantic validation and feedback/error flows
Bharath Jeyakkumar S	Automated testing and verification
Prasannavelan K	SQLAlchemy persistence, setup and demonstration support